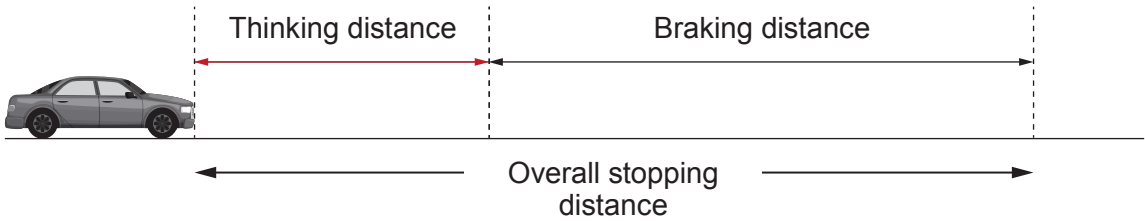
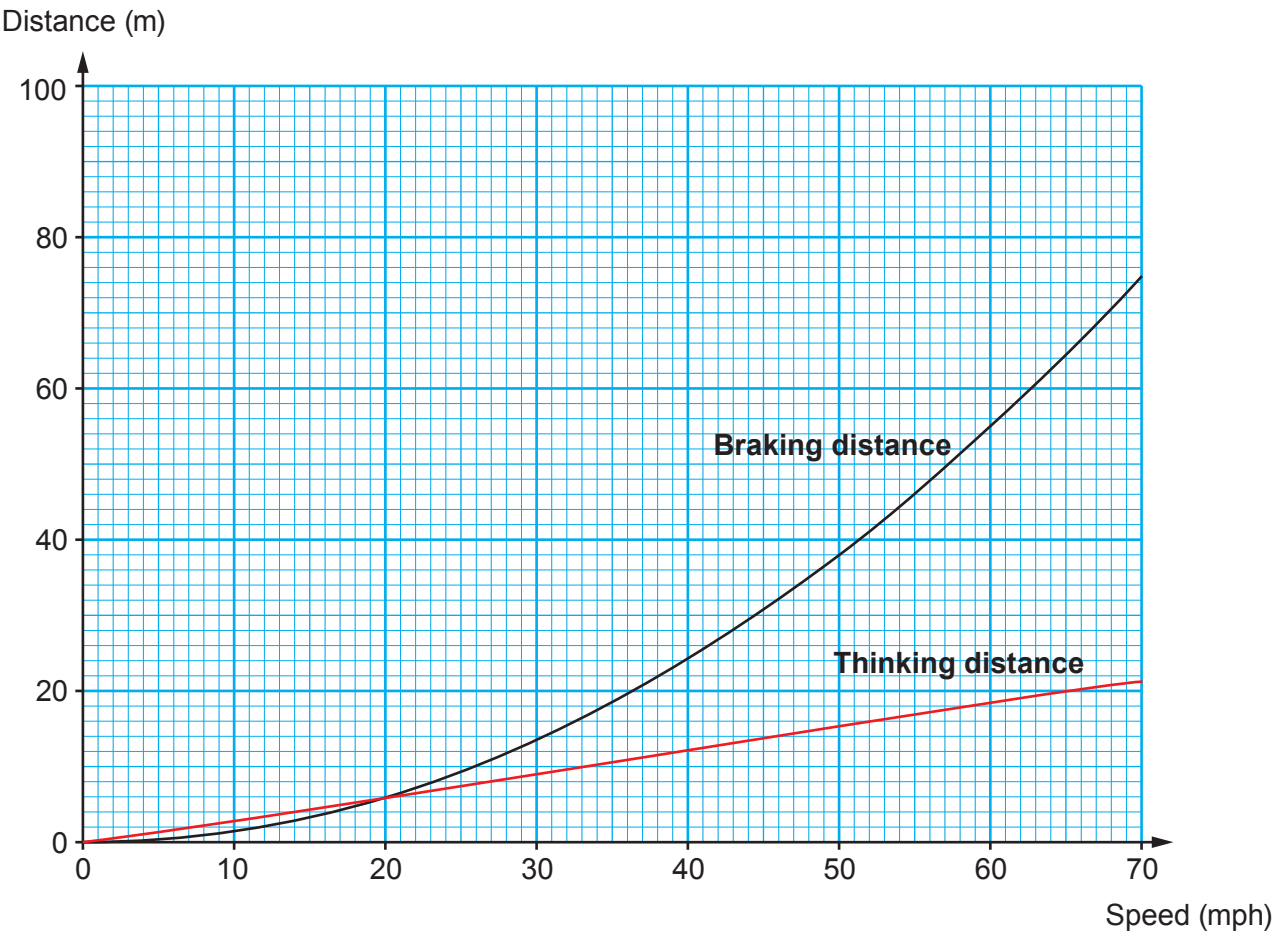


Answer all questions.

1. The overall stopping distance of a car is made up of two parts:
- the distance that the car travels when the driver is reacting (thinking distance)
  - the distance that the car travels after the brakes have been applied (braking distance).



The graph below shows how the thinking distance and braking distance depend on the speed of a vehicle under good conditions.



The table below shows the conversion from mph into m/s.

|             |    |    |    |    |
|-------------|----|----|----|----|
| Speed (mph) | 20 | 40 | 60 | 70 |
| Speed (m/s) | 9  | 18 | 27 | 31 |



- (a) (i) It is suggested that both thinking distance and braking distance are directly proportional to speed. Explain whether this suggestion is true. [2]

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- (ii) Use information on page 4 and the equation:
- $$\text{time} = \frac{\text{distance}}{\text{speed}}$$
- to calculate the **thinking** time of the driver when travelling at 40 mph. [3]

Thinking time = ..... s

- (iii) Use the information on the graph to complete the table below. [2]

|                               |       |       |       |       |       |       |
|-------------------------------|-------|-------|-------|-------|-------|-------|
| Speed (mph)                   | 0     | 20    | 30    | 40    | 60    | 70    |
| Overall stopping distance (m) | ..... | ..... | ..... | ..... | ..... | ..... |

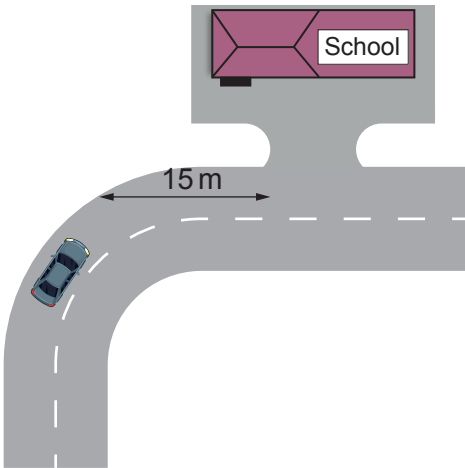
- (iv) Use the data in the table to **plot** the points on the grid opposite **and draw a line** to show how the overall stopping distance depends on speed. [3]

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(b) The speed limit along a road outside a school in Cardiff was 30 mph. The council decided to reduce this to 20 mph in 2017.

The entrance to the school is situated 15 m after a bend in the road.



Explain how the change in speed limit affects the chance of children getting knocked down as they cross the road outside the school entrance. Use data to support your answer. [3]

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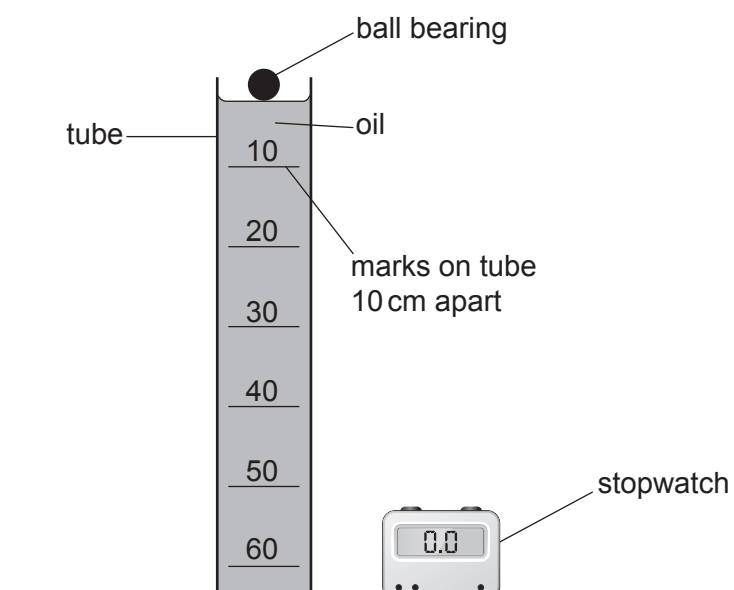
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2. Students investigate the terminal speed of a ball bearing in oil. They measure the time it takes for the ball bearing to drop different distances through the oil.



The results from the experiment are shown in the table below.

| Distance (cm) | Time (s) |         |         |      |
|---------------|----------|---------|---------|------|
|               | Trial 1  | Trial 2 | Trial 3 | Mean |
| 10            | 5.4      | 6.2     | 5.8     | 5.8  |
| 20            | 7.6      | 4.2     | 8.0     | 7.8  |
| 30            | 8.4      | 9.0     | 8.3     | 8.6  |
| 40            | 10.9     | 10.3    | 10.4    | 10.5 |
| 50            | 11.5     | 11.2    | 10.8    | 11.2 |
| 60            | 12.5     | 13.0    | 13.2    | 12.9 |

- (a) (i) Freya states that the mean time for 20 cm is incorrect and should be 6.6 s. Explain whether Freya is correct. [2]

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(ii) Evaluate the repeatability of the data.

[1]

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(b) State **one** source of inaccuracy in this method **and** how it could be reduced.

[2]

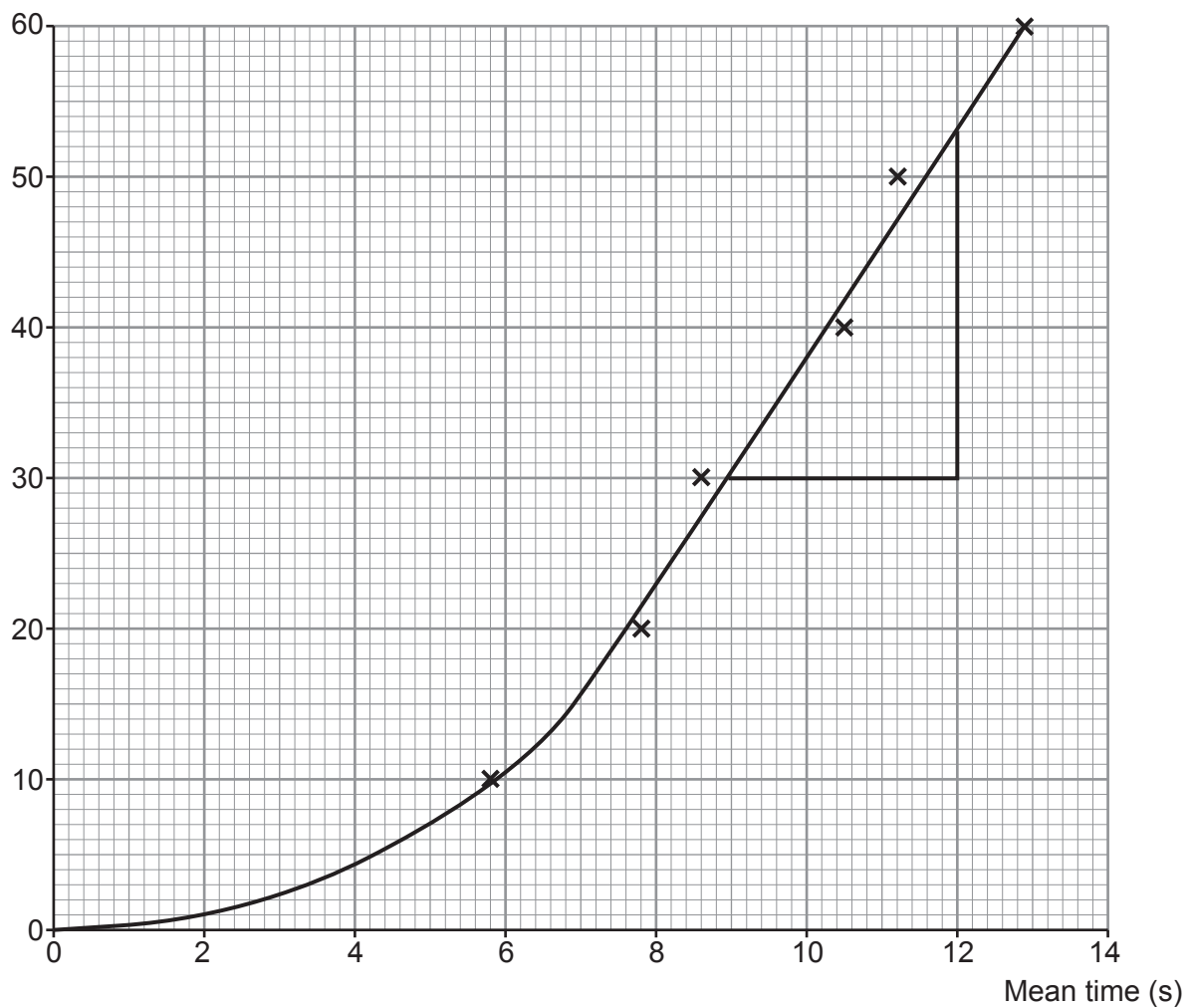
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(c) A graph of the students' data is given below.

Distance (cm)



The section of the graph where the line is straight represents the ball bearing travelling at terminal speed.

- (i) Estimate the time at which the ball bearing reaches terminal speed. [1]

time = ..... s

- (ii) The gradient of a distance-time graph represents the speed of an object.  
Use the equation:

speed = gradient of distance-time graph

and the triangle shown on the graph to calculate the terminal speed of the ball bearing. [2]

terminal speed = ..... cm/s

- (iii) State how the **acceleration** of the ball bearing changes as it falls through the oil. [2]

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